

ISA 444/544: Business Forecasting

04: Plotting a Single Time Series in Python (Cont.)

Fadel M. Megahed, PhD

Raymond E. Gos Professor in Business
Farmer School of Business
Miami University

 @FadelMegahed

 fmegahed

 fmegahed@miamioh.edu

 Automated Scheduler for Office Hours

Fall 2025

Quick Refresher of Last Class

- ✓ Generate and interpret simple line charts
- ✓ Create seasonal plots and subplots

Learning Objectives for Today's Class

- Describe a lag and create lag scatter plots.
- Plot and interpret the autocorrelation function (ACF) for a time-series.

Describe a lag and create lag scatter plots

What is a Lag?

A **lag** refers to a time delay between an event and its observable effect or measurement.

In time series data, a **lag** is simply a previous observation in the series. For example:

- **Lag 1:** Refers to the value immediately preceding the current value.
- **Lag 2:** Refers to the value two time periods before the current value, and so on.

What is a Lag (Cont.)?

Imagine a time series of daily sales:

- Day 1: \$100
- Day 2: \$150
- Day 3: \$120

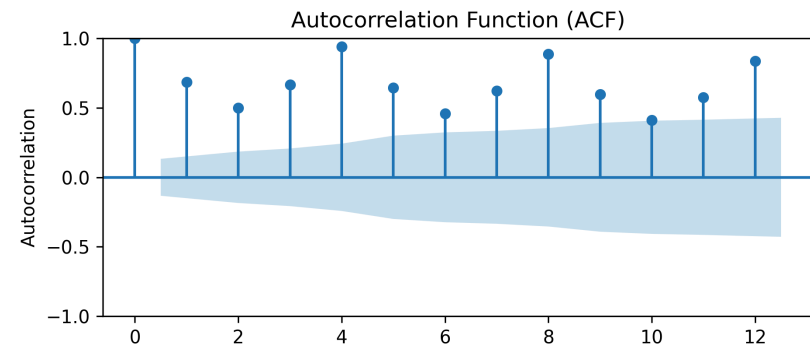
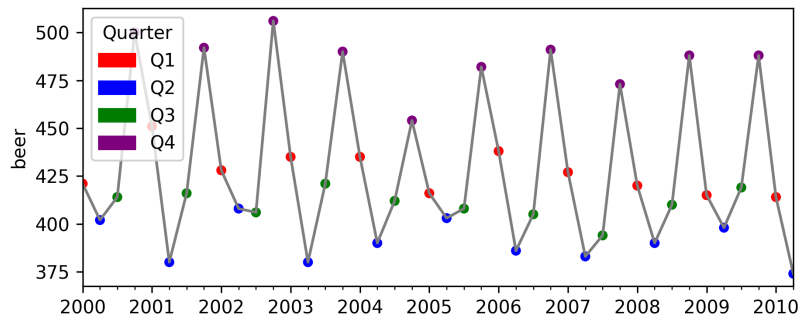
For Day 3, the lag 1 value is the sales from Day 2 (\$150). Similarly, the lag 2 value is from Day 1 (\$100).

In mathematical notation, y_t represents the value at time t , then:

- y_{t-1} is the lag 1 value of y_t
- y_{t-2} is the lag 2 value of y_t

Importance of Lags

- **Lags** are important in time series analysis because they can help us understand the relationship between past and present values. They are used in **Forecasting**:
 - **Autoregressive models** Models such as AR (AutoRegressive) and ARIMA models use past values (lags) of the series to predict future values. For example, an AR(1) model uses predicts y_t solely based on y_{t-1} .
 - **Seasonality**: Lags help capture seasonal patterns in data.



Importance of Lags

- **Lags** are important in time series analysis because they can help us understand the relationship between past and present values. Hence, they are used in:
 - **Causal Analysis:**
 - **Cause and Effect:** In many business scenarios, past events (lags) can be predictors for current outcomes; e.g., past advertising spend might affect current sales.
 - **Policy Evaluation:** Understanding the delayed impact of policies or changes (e.g., price changes, marketing campaigns) is essential.

Importance of Lags (Cont.)

- **Lags** are important in time series analysis because they can help us understand the relationship between past and present values. Hence, they are used in:
 - **Data Preparation and Feature Engineering:**
 - **Feature Creation:** In machine learning models, creating lag features from time series data can help the model understand historical patterns and trends.
 - **Smoothing and Differencing:** Lags are also used in techniques like differencing to stabilize the mean of a time series, which is a common preprocessing step.

Implementing Lags in Python

In Python, we can create lags using the `shift()` method in pandas.

```
import pandas as pd

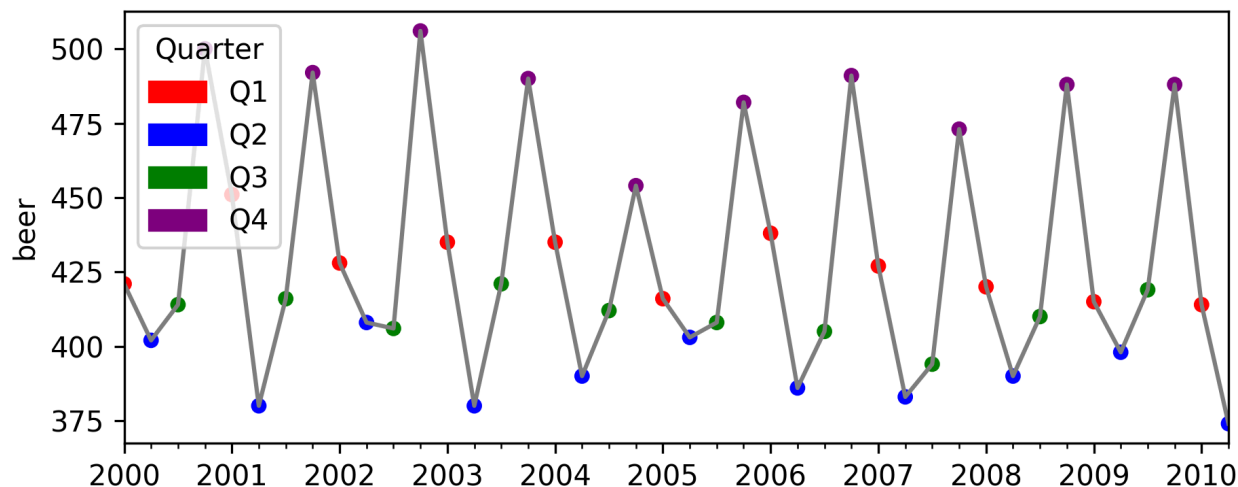
df = (
    pd.DataFrame(
        {
            'date': pd.date_range(start='2025-01-01', periods=5, freq='D'),
            'sales': [100, 150, 120, 200, 180]
        }
    )
    .assign(
        lag1 = lambda x: x['sales'].shift(1),
        lag2 = lambda x: x['sales'].shift(2)
    )
)

df.head(3)
```

##		date	sales	lag1	lag2
## 0		2025-01-01	100	NaN	NaN
## 1		2025-01-02	150	100.0	NaN
## 2		2025-01-03	120	150.0	100.0

Class Activity: Plot the aus_beer Time Series

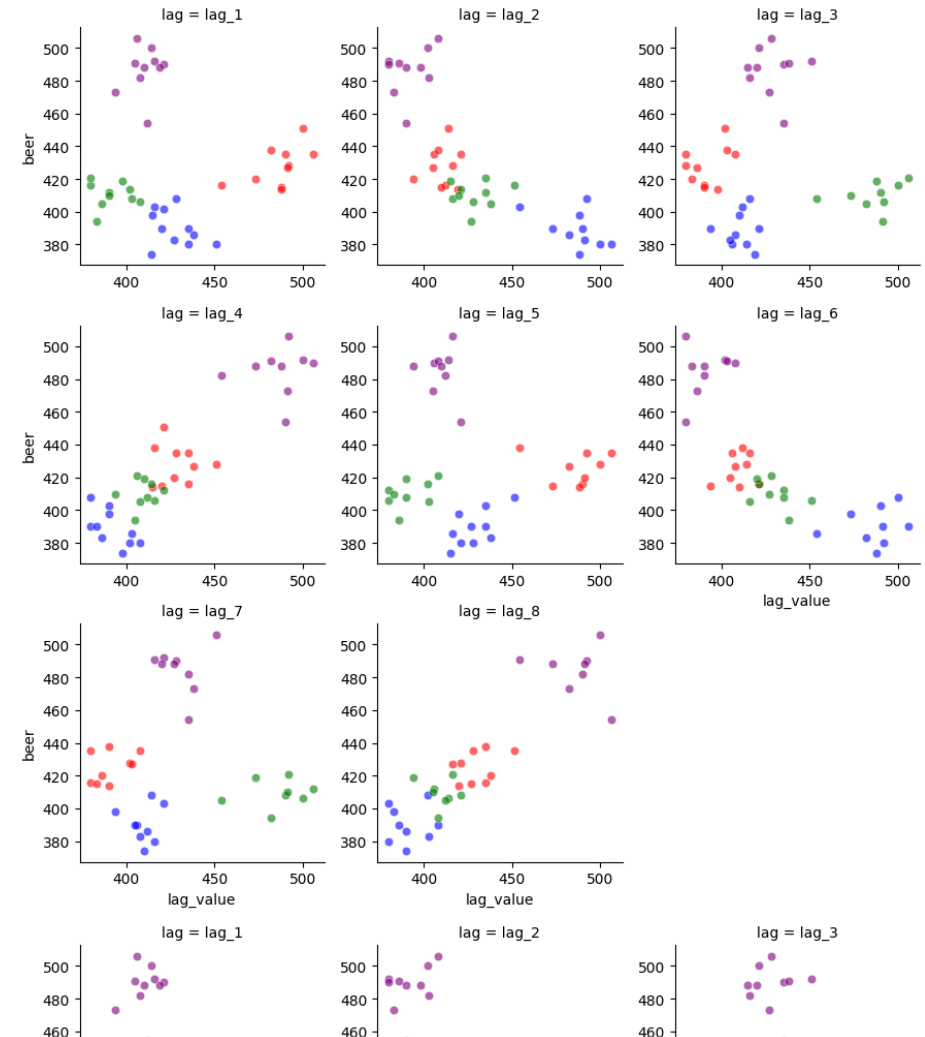
- Load the `aus_beer.csv` dataset from Canvas.
- Create a line plot of the `beer` column.
- Color the plot by quarter.
- Add a legend to the plot.



Demo: Creating and Interpreting Lag Scatter Plots

In this demo, we will:

- Create lag scatter plots for a time series.
- Interpret the relationship between the current value and its lagged values.



Plot and Interpret the ACF for a Time Series

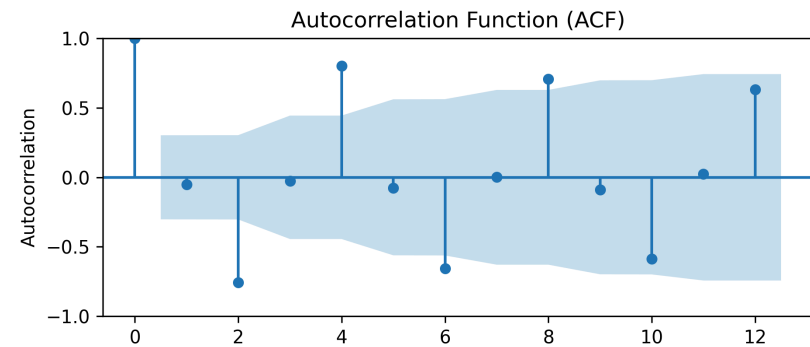
What is the Autocorrelation Function (ACF)?

The **Autocorrelation Function (ACF)** is a measure of the correlation between a time series and a lagged version of itself.

```
from statsmodels.graphics.tsaplots import
import matplotlib.pyplot as plt

df = pd.read_csv('../data/aus_beer.csv')

plot_acf(df['beer'], lags=12)
plt.title('Autocorrelation Function (ACF)')
plt.xlabel('Lag (quarter)')
plt.ylabel('Autocorrelation')
plt.show()
```



Recap

Summary of Main Points

By now, you should be able to do the following:

- Describe a lag and create lag scatter plots.
- Plot and interpret the ACF of a time-series.



Review and Clarification



- **Class Notes:** Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording:** The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Questions:** Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.